

# When Frozen Multimodal Hallucination Wrappers Fail

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## Abstract

Training-free wrappers promise a cheap way to reduce hallucination in frozen multimodal language models, yet their scored-output behavior can be indistinguishable from the baseline they wrap. We audit calibration-free wrapper families on selected 300-row POPE, AMBER, and HallusionBench subsets with a frozen Qwen/Qwen2.5-VL-7B-Instruct backend and official scorer outputs. The retained matrix does not support a positive mitigation claim: prompt-label verifier reranking and evidence-companion arbitration are base-identical on normalized official answers, NoLan fixed-alpha decoding is non-decisive and parser-sensitive on HallusionBench, and visual-delta answer-token contrast degrades AMBER by -6.3 points with McNemar  $p = 0.016348$  and CI [-0.112, -0.014]. HallusionBench visual-delta also decreases by -3.7 points, but its paired interval [-0.088, +0.014] spans zero. These results frame frozen wrappers as diagnostic probes whose failures expose boundary conditions for future decoding-time and evidence-grounded hallucination work.

## Introduction

Multimodal hallucination work has moved from caption-only object errors toward object, attribute, relation, and visual-question failures in large vision-language models. Early captioning analyses established that models can name objects absent from an image (Rohrbach et al. 2018), and modern evaluation suites such as POPE, AMBER, HallusionBench, THRONE, triplet-level evaluation, and Reefknot broaden the target to discriminative object existence, relation, and entangled visual-language reasoning (Li et al. 2023; Wang et al. 2023; Guan et al. 2024; Kaul et al. 2024; Wu et al. 2024; Zheng et al. 2024). The practical pressure is clear: a frozen deployed model is easier to wrap than to re-train, so a large recent line studies decoding-time, attention, visual-contrast, grounding, and correction mechanisms that aim to suppress hallucinated answers without changing the base model weights (Leng et al. 2024; Huang et al. 2024; Ren et al. 2026; Zhao et al. 2024; He et al. 2025; Li et al. 2025; Deng and Yang 2025).

That attraction creates an evaluation trap. A wrapper can add a verifier prompt, a contrastive score, or an evidence companion around a frozen backend while still leaving retained official answers unchanged, or it can change many answers in a direction that hurts one benchmark and

only weakly helps another. Recent 2026 proposals sharpen this problem because many are explicitly training-free or calibration-light and are motivated by language-prior reduction, token-level visual sensitivity, or tool-free attribution (Anand et al. 2026; Li et al. 2026b; Qin et al. 2026; Tang et al. 2026; Lei et al. 2026; Guo et al. 2026; Zhang et al. 2026; Tan et al. 2026). A negative audit of such wrappers is therefore useful only if it separates three states: no retained answer change, non-decisive paired movement, and decisive degradation.

The separation is especially important because the newest intervention families differ in what they claim to perturb. Some target visual tokens or localized attention, some steer heads or reduce language-prior dominance, and others introduce attribution, caption guidance, or causal decoding objectives (Ogunleye, Abdelrahman, and Lourentzou 2026; Yao et al. 2026; Cheng et al. 2026b; Chang et al. 2026; Huang, Zhang, and Luo 2026; Yu et al. 2026). A mean-score table alone cannot tell whether such mechanisms created a legitimate visual correction, a harmless formatting change, or a systematic answer-token bias. Our audit therefore treats answer identity as a first-class outcome. If the official-answer string is unchanged after normalization, the wrapper is counted as a no-op for the retained scorer even when its intermediate controller output differs. If an answer changes, the row is classified by whether the change fixes or harms paired correctness.

This paper asks a deliberately narrow question: under a retained public-benchmark matrix, when do calibration-free wrappers around a frozen multimodal model produce real official-answer changes, and when do they merely look like mitigation machinery? Figure 1 summarizes the diagnostic design. We compare four wrapper probes against a frozen prompt baseline on three benchmark families, using official scorer outputs and paired row tests rather than unpaired score summaries. The contribution is not a new positive mitigation method. It is a failure-mode audit with three findings. First, prompt-label verifier reranking and evidence-companion arbitration are normalized-answer identical to the frozen baseline across POPE, AMBER, and HallusionBench. Second, NoLan fixed-alpha decoding has small retained-parser deltas on POPE (+0.7 points), AMBER (+0.3 points), and HallusionBench (+1.7 points), all intervals span zero, and a strict invalid-as-wrong sen-

sitivity changes the HallusionBench NoLan delta to -1.0 points. Third, visual-delta answer-token contrast changes many rows and significantly hurts AMBER, while its HallusionBench decrease is not decisive. These findings argue for retaining row-level identity, parser sensitivity, and harm diagnostics whenever wrapper papers report training-free hallucination gains.

The paper is scoped as an evidence audit rather than an algorithm release. It makes one reproducible diagnostic point: wrapper plausibility does not equal retained scorer improvement. The resulting evidence standard is concrete. A calibration-free wrapper should produce non-identity where its mechanism is supposed to act, beneficial rather than harmful paired changes, and scorer agreement strong enough to survive parser scrutiny.

## Related Work

Hallucination evaluation has become more fine-grained than single caption-level object checks. POPE evaluates yes/no object-existence consistency from image-question pairs (Li et al. 2023), AMBER adds an LLM-free multidimensional benchmark for hallucination assessment (Wang et al. 2023), and HallusionBench targets entangled visual illusion and language hallucination in question answering (Guan et al. 2024). Broader benchmark work studies free-form objects, triplets, relations, clinical reasoning, and modality-agnostic definitions (Kaul et al. 2024; Wu et al. 2024; Zheng et al. 2024; Yang et al. 2026; Hossan, Tong, and Deng 2026). Our audit follows this benchmark-centered tradition but keeps the claim narrower: official scorer outputs and row-paired changes define the evidence boundary.

Training-free or lightweight interventions often target decoding, visual attention, or post-hoc correction rather than full retraining. Visual contrastive decoding and contrastive hallucination visualization use differences between visual and linguistic signals to steer answers (Leng et al. 2024; Park et al. 2025); OPERA penalizes over-trust and allocates retrospection (Huang et al. 2024); RFI and ASCD intervene in flow or attention-steerable contrastive decoding (Cheng et al. 2026a; Wang et al. 2026). Recent proposals also explore language-prior suppression, image-head masking, stability-aware knowledge, visual token engineering, context-aware attention, token-level steering, and caption-guided attention steering (Ren et al. 2026; Deng and Yang 2025; Li et al. 2026b; Tang et al. 2026; Lei et al. 2026; Zhang et al. 2026; Li et al. 2026a). These papers motivate our design-space comparison, but our retained evidence rejects a positive wrapper claim.

Another line uses extra grounding, segmentation, counterfactuals, or attribution to detect or correct hallucinations. Woodpecker and image-grounded guidance add correction or grounding modules (Yin et al. 2023; Zhao et al. 2024); newer work studies segmentation-before-answering, propose-and-attend confidence, counterfactual video grounding, and shared-prefix attribution (Wei et al. 2026; Shalam et al. 2026; Poppi et al. 2026; Qin et al. 2026). The gap is that a wrapper can be mechanistically plausible while leaving the retained output distribution unchanged or harming a discriminative benchmark. We therefore report

identity rates, harmful changed rows, and exact paired tests next to mean scores.

This framing also differs from benchmark papers that primarily expand the taxonomy of hallucination. Those papers supply the measurement targets; here the focus is on wrapper failure semantics once a public scorer is fixed. For example, a relation-oriented benchmark may expose different errors than an object-existence benchmark, but a training-free wrapper still has to move the retained answer in the correct direction under that benchmark’s scorer. Our related-work dependency is therefore asymmetric: prior work motivates the wrapper families and benchmark families, while row-level scorer outputs determine whether the evaluated wrapper actually changes official answers. This prevents recent method names from becoming unsupported claims in the manuscript.

## Diagnostic Design Space

We treat each wrapper as an external decision policy around a frozen visual-question answering backend. The frozen prompt baseline queries Qwen/Qwen2.5-VL-7B-Instruct with a yes/no answer format, and every method is judged after the same benchmark-side normalization. No wrapper is trained, tuned on a calibration set, or allowed to update model parameters. The design question is whether the wrapper changes retained official answers in the right direction, not whether the surrounding mechanism appears plausible.

The episode interface is the same across methods. Each retained task contains an image-conditioned binary question, a benchmark-side gold label, a raw answer emitted by the frozen backend or wrapper, and a normalized yes/no answer used for retained scoring. The baseline decodes directly from the frozen prompt. A wrapper may add extra candidate generation, labels, contrast signals, or companion evidence before the final answer is normalized. The analysis then joins wrapper rows to their baseline rows by benchmark family and task identifier. This join is what permits McNemar tests, identity rates, and beneficial/harmful changed-row counts. Without that join, a method could report a mean delta while hiding whether it changed the same items as the baseline.

We distinguish three increasingly strict notions of change. Exact identity means the raw answer string is unchanged. Normalized identity means the raw strings may differ, but the retained binary answer is the same; this matters because case and formatting changes should not count as hallucination mitigation. Paired correctness change means the normalized answer differs and the retained gold label determines whether the wrapper fixed or broke the row. The main claims use normalized identity and paired correctness, not raw text churn. This choice is conservative for wrappers that produce explanatory text, because an explanation is not credited unless it changes the retained official answer correctly.

NoLan fixed-alpha decoding is the only imported named wrapper in the retained matrix. It is motivated by suppressing language priors in large vision-language models

# Frozen Wrapper Audit

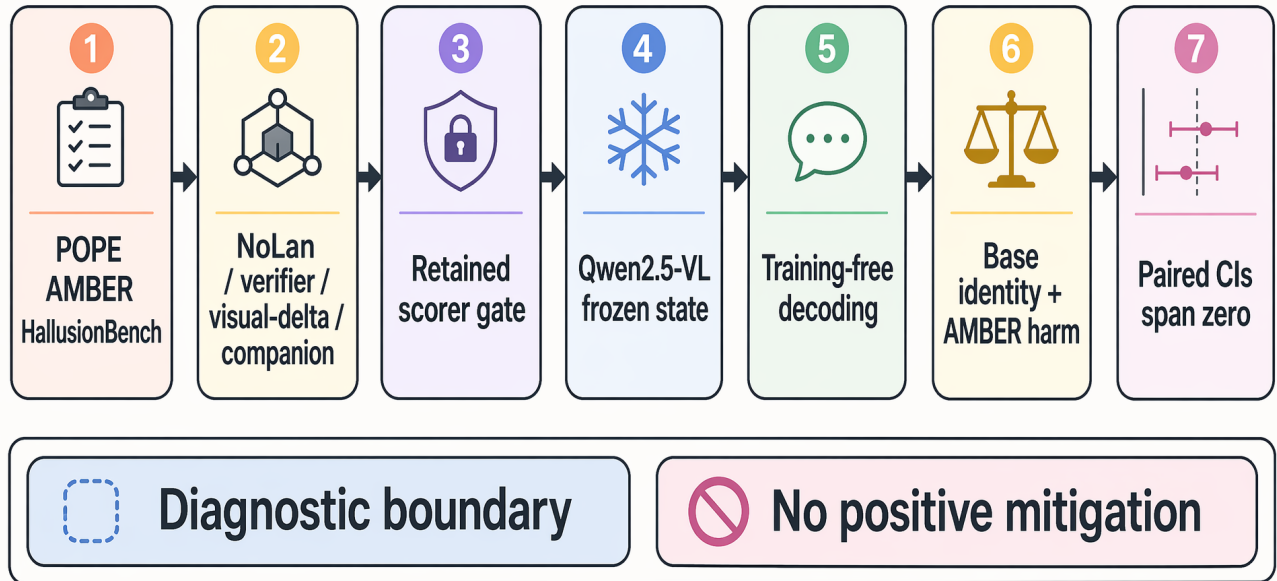


Figure 1: Frozen-wrapper audit for training-free hallucination probes. The retained matrix separates no-op, non-decisive, and harmful behavior rather than supporting a positive mitigation claim.

(Ren et al. 2026), but here the evaluated setting is a fixed-alpha, calibration-free variant rather than a tuned full reproduction. Its paper-facing question is whether a simple prior-suppression wrapper creates beneficial paired answer changes without introducing harms. The retained evidence shows some movement: normalized identity is 98.0% on POPE, 95.7% on AMBER, and 93.0% on HallusionBench, but the corresponding confidence intervals span zero.

The prompt-label verifier rerank probe runs one baseline answer call and seven fixed 16-token forced-choice probes per row. The probes cover object, attribute, relation, companion consistency, yes support, no support, and conflict direction; each probe must return exactly one allowed label. The wrapper preserves the baseline answer unless all probes are valid, the conflict slot requests a direction change, the yes/no support slots agree with that direction, and the object, attribute, relation, or companion slot supplies compatible evidence. This is a reranking-style controller because it can only choose between the baseline yes/no decision and the opposite binary answer; it uses no gold labels, no learned threshold, and no calibration set.

The evidence-companion arbitration probe runs one baseline answer call and one 128-token evidence-ledger call per row. The ledger must fill fixed fields for visual anchor, attribute evidence, relation evidence, answer support, answer conflict, conflict type, and rationale. The controller changes the answer only when the ledger is valid, the conflict type is one of the allowed answer-changing types, the support and conflict fields agree with the proposed direction, and

companion-inconsistency claims have complete companion evidence. In the retained matrix, evidence conflict and companion completeness do not form a usable arbitration channel, so the wrapper preserves the baseline answer.

The visual-delta answer-token contrast probe tests a more aggressive hypothesis inspired by contrastive decoding. It first obtains the baseline answer, then computes next-token log-probabilities for canonical yes and no under an image-conditioned pass and a text-only matched-question pass. The selected binary answer is the token with the larger visual delta, defined as image log-probability minus text-only log-probability; invalid tokenization, invalid passes, non-finite scores, or exact ties preserve the baseline answer. This wrapper is the only one that creates many changed rows on every benchmark. On AMBER, that activity lowers all-accuracy by -6.3 points with a confidence interval below zero; on HallusionBench, the decrease is -3.7 points but the interval spans zero.

The design-space audit therefore has two controls around the aggressive visual-delta probe. The no-op controls show that wrapper scaffolding alone cannot justify a mitigation claim. NoLan shows a small-movement condition where a named training-free prior-suppression idea changes some rows but not enough to support a decisive retained effect. Visual-delta shows a high-movement condition where a plausible contrastive signal can move answers in the wrong direction. Together, these controls identify which failure mode occurs for each wrapper family.

Table 1: Three public benchmark families contribute 900 baseline rows and 4,500 scored method rows.

| Benchmark      | Rows | Base score |
|----------------|------|------------|
| POPE           | 300  | 88.7%      |
| AMBER          | 300  | 89.3%      |
| HallusionBench | 300  | 84.3%      |

POPE uses COCO-random retained accuracy; AMBER uses existence/attribute/relation all-accuracy; HallusionBench uses question accuracy. Each family has five method cells.

## Experimental Setup

The retained matrix uses three public benchmark families. POPE contributes a 300-row COCO random object-existence yes/no subset scored with POPE-style retained accuracy (Li et al. 2023). AMBER contributes a 300-row discriminative existence/attribute/relation subset scored with retained all-accuracy (Wang et al. 2023). HallusionBench contributes 300 visual-question rows scored at retained question level (Guan et al. 2024). Each method-by-benchmark cell contains one retained 300-row run, for 15 cells and 4,500 scored rows overall. The method comparisons use 12 wrapper-versus-baseline paired records, totaling 3,600 paired rows.

Table 1 gives the benchmark-facing contract. The selected slices are intentionally balanced in row count so each family contributes the same number of retained questions to the wrapper audit. They are not interchangeable, however. POPE is the simplest binary object-existence setting. AMBER mixes discriminative object, attribute, and relation phenomena and reports retained all-accuracy. HallusionBench stresses visual-language illusion and question-level consistency; the retained subset does not contain the original companion rows needed for pair-all decomposition. This is why HallusionBench claims in the paper stay at question-level paired correctness.

The evaluated backend is Qwen/Qwen2.5-VL-7B-Instruct in a frozen yes/no setting. All generated-answer paths use deterministic decoding with temperature 0, no sampling, and at most 128 new tokens. The frozen prompt baseline makes one image-conditioned answer call. NoLan fixed-alpha generates each token from the fixed formula  $(1 + \alpha)\ell_{image} - \alpha\ell_{text}$  with  $\alpha = 1.0$  for up to 128 tokens. Prompt-label verifier reranking uses one baseline call plus seven fixed label probes capped at 16 tokens each. Visual-delta contrast uses one baseline call plus image-conditioned and text-only next-token score passes for canonical yes/no tokens. Evidence-companion arbitration uses one baseline call plus one fixed evidence-ledger call capped at 128 tokens. All controllers make deterministic rule-based decisions and do not use benchmark gold labels.

The paper-facing metric is official-score accuracy for POPE, all-accuracy for AMBER, and question accuracy for HallusionBench. For wrapper comparisons, we compute exact McNemar tests on paired binary row correctness, 95 percent confidence intervals on score deltas, normalized identity rates, and counts of beneficial and harmful changed rows. The comparison unit is the paired task row: a wrap-

per either leaves the normalized official answer unchanged, changes a small number of paired answers with uncertainty spanning zero, or changes enough answers in the wrong direction to produce a significant degradation. This row-level unit keeps benchmark-specific scorer outputs, identity rates, and harm counts aligned without pooling the three sources into one leaderboard.

Several boundaries matter for interpretation. HallusionBench pair-all decomposition is unavailable because the selected subset lacks the required original companion rows, so HallusionBench paired claims are question-level. The yes/no parser first checks explicit negative and affirmative markers, then maps unrecognized binary text to yes; therefore we additionally run a strict invalid-as-wrong sensitivity check over raw predictions. It finds one parser-sensitive cell: HallusionBench NoLan has 9/300 non-binary raw outputs, changing its score from 86.0% retained accuracy to 83.3% when invalid raw answers are counted wrong.

We also avoid treating the three benchmark families as interchangeable samples from one pooled population. POPE provides a relatively direct object-existence stress test, AMBER mixes object, attribute, and relation phenomena inside a retained all-accuracy score, and HallusionBench contributes question-level visual-language illusion rows without the companion structure needed for pair-all decomposition. The setup is consequently designed for comparative failure semantics rather than a single pooled leaderboard. If a wrapper succeeds on one family and fails on another, the family-specific paired evidence is the result, not a nuisance to be averaged away.

## Results

Table 2 is the central result. Each row reports source, task split, metric, backend, call policy, score, and delta. Prompt-label verifier reranking and evidence-companion arbitration match the frozen baseline on every retained mean; Table 3 confirms zero normalized changed rows for both probes.

The matrix separates output identity from uncertain movement. Equal scores establish a no-op only when paired rows show no normalized answer changes; a nonzero delta is informative only when uncertainty, changed-row direction, and parser sensitivity agree. NoLan moves too little for a decisive claim, whereas visual-delta moves more often but is harmful on AMBER and non-decisive on HallusionBench.

NoLan fixed-alpha has small deltas whose paired intervals all include zero. Counting its HallusionBench non-binary outputs as wrong also reverses that cell’s direction. We therefore report limited, parser-sensitive movement rather than cross-benchmark mitigation.

Visual-delta is the sharpest negative finding: POPE moves upward within uncertainty, AMBER shows the decisive harm in Table 3, and HallusionBench moves downward within uncertainty.

The cross-benchmark pattern rules out a simple “more visual contrast is better” story. POPE is compatible with a possible benefit, AMBER shows a significant degradation, and HallusionBench moves downward without decisive paired evidence. The AMBER type split localizes the aggregate

Table 2: Across 4,500 scored rows, the matrix shows two no-op wrappers, non-decisive NoLan movement, and -6.3 AMBER visual-delta harm.

| Benchmark   | <i>N</i> | Backend       | Method / role | Decoding / budget | Metric   | Score       | $\Delta$ |
|-------------|----------|---------------|---------------|-------------------|----------|-------------|----------|
| POPE        | 300      | Qwen2.5-VL-7B | Base          | greedy/128        | Acc.     | 88.7        | 0.0      |
|             |          |               | NoLan         | alpha=1/128       |          | 89.3        | +0.7     |
|             |          |               | Verifier      | 7 probes          |          | 88.7        | 0.0      |
|             |          |               | V-delta       | img-text Y/N      |          | <b>91.7</b> | +3.0     |
|             |          |               | Companion     | ledger/128        |          | 88.7        | 0.0      |
| AMBER       | 300      | Qwen2.5-VL-7B | Base          | greedy/128        | All-acc. | 89.3        | 0.0      |
|             |          |               | NoLan         | alpha=1/128       |          | <b>89.7</b> | +0.3     |
|             |          |               | Verifier      | 7 probes          |          | 89.3        | 0.0      |
|             |          |               | V-delta       | img-text Y/N      |          | 83.0        | -6.3     |
|             |          |               | Companion     | ledger/128        |          | 89.3        | 0.0      |
| HallusionB. | 300      | Qwen2.5-VL-7B | Base          | greedy/128        | Q-acc.   | 84.3        | 0.0      |
|             |          |               | NoLan         | alpha=1/128       |          | <b>86.0</b> | +1.7     |
|             |          |               | Verifier      | 7 probes          |          | 84.3        | 0.0      |
|             |          |               | V-delta       | img-text Y/N      |          | 80.7        | -3.7     |
|             |          |               | Companion     | ledger/128        |          | 84.3        | 0.0      |

Blank source cells inherit the benchmark block’s *N*, Qwen2.5-VL-7B backend, and metric. Method and budget remain explicit per row; Table 3 reports paired uncertainty.

harm to existence and relation rather than attribute questions: attribute remains 87.0% to 87.0% (8 beneficial and 8 harmful changes), existence falls from 96.0% to 87.0% (0 beneficial and 9 harmful), and relation falls from 85.0% to 75.0% (11 beneficial and 21 harmful). This typed diagnosis is why the result is reported as a failure mode rather than as an ablation of a successful method.

The evidence forms two diagnostic stages. The first stage asks whether a wrapper changes retained answers at all. Prompt-label verifier and evidence companion fail here, so their identical mean scores are meaningful no-op evidence rather than inconclusive score ties. The second stage asks whether changed answers move in the correct direction. NoLan and visual-delta reach this stage, but NoLan remains within paired uncertainty while visual-delta fails directionally on AMBER. This two-stage interpretation is the core diagnostic contribution: mean score, normalized identity, beneficial and harmful changed rows, and paired uncertainty need to appear together.

## Mechanism Analysis and Ablation

Table 3 adds the row-paired layer behind the mean matrix. The two 100% identity rows are controls: their controller policies execute, but they do not cross the retained answer boundary. This makes the no-op outcome interpretable rather than an ordinary score tie.

The two controls fail at different internal gates. Prompt-label verifier reranking adds semantic labels and conflict checks without changing official answers. Evidence-companion arbitration adds a ledger, but the retained ledger fields do not activate a usable conflict channel. In both cases, extra wrapper structure remains outside the scored decision.

NoLan fixed-alpha decoding behaves differently. It changes a small number of normalized answers and sometimes fixes errors, but it also introduces harms. These are the right kinds of changes for a possible mitigation, but the paired intervals remain too wide for the retained matrix to

support a positive result. The appropriate wording is therefore “non-decisive retained movement,” not “improvement.”

Visual-delta contrast shows why row-level harm counts are necessary for training-free wrappers. The AMBER result is not just a lower aggregate score; it is a paired degradation. HallusionBench has similar directional pressure but less decisive paired evidence, so we report it as an observed retained-run decrease whose interval spans zero. The same scoring and row-join procedure is used for every method cell. The remaining interpretation is methodological: answer-token visual contrast can move labels aggressively, and those movements can be harmful under retained official scoring.

The failure taxonomy gives the mechanism-level reading. Evidence companion fails because its ledger mostly does not produce a decision-changing conflict signal. Prompt-label verifier fails because diagnostic labels do not translate into changed normalized answers. Visual-delta fails differently: it changes many normalized answers, so its mechanism is active, but the direction of change is unsafe on AMBER and only weakly supported elsewhere. NoLan sits between these extremes: it changes 40 of 900 paired rows, but the beneficial and harmful counts remain too small for decisive cross-family support. Collapsing these outcomes into “wrappers do not work” would be too broad; reporting them separately makes the negative result useful for designing the next benchmark or wrapper.

The paired records also expose distinct repair targets. The verifier and companion each preserve all 900 normalized outputs, so a successor must first demonstrate a decision-changing channel. NoLan changes 40 rows, which motivates wider sampling before a directional claim. Visual-delta changes 155 rows, but AMBER contains 38 harms against 19 benefits. Its problem is therefore unsafe direction rather than inactivity. Separating these cases prevents no-op and harmful mechanisms from being treated as the same negative result.

Table 3: AMBER visual-delta is harmful ( $p = 0.016348$ , CI  $[-0.112, -0.014]$ ); other shown nonzero intervals include zero.

| Benchmark      | Wrapper               | $\Delta$ | McNemar $p$ | 95% CI             | Id.   | Benefit | Harm |
|----------------|-----------------------|----------|-------------|--------------------|-------|---------|------|
| POPE           | NoLan fixed-alpha     | +0.7     | 0.687500    | $[-0.009, +0.023]$ | 98.0  | 4       | 2    |
| AMBER          | NoLan fixed-alpha     | +0.3     | 1.000000    | $[-0.020, +0.027]$ | 95.7  | 7       | 6    |
| HallusionBench | NoLan fixed-alpha     | +1.7     | 0.383310    | $[-0.013, +0.047]$ | 93.0  | 13      | 8    |
| POPE           | Prompt-label verifier | 0.0      | 1.000000    | $[0.000, 0.000]$   | 100.0 | 0       | 0    |
| AMBER          | Prompt-label verifier | 0.0      | 1.000000    | $[0.000, 0.000]$   | 100.0 | 0       | 0    |
| HallusionBench | Prompt-label verifier | 0.0      | 1.000000    | $[0.000, 0.000]$   | 100.0 | 0       | 0    |
| AMBER          | Visual-delta contrast | -6.3     | 0.016348    | $[-0.112, -0.014]$ | 81.0  | 19      | 38   |
| HallusionBench | Visual-delta contrast | -3.7     | 0.200031    | $[-0.088, +0.014]$ | 79.7  | 25      | 36   |
| POPE           | Evidence companion    | 0.0      | 1.000000    | $[0.000, 0.000]$   | 100.0 | 0       | 0    |
| AMBER          | Evidence companion    | 0.0      | 1.000000    | $[0.000, 0.000]$   | 100.0 | 0       | 0    |
| HallusionBench | Evidence companion    | 0.0      | 1.000000    | $[0.000, 0.000]$   | 100.0 | 0       | 0    |

Id. is normalized identity. Benefit/Harm count wrong-to-correct and correct-to-wrong changed rows;  $p$  is exact McNemar.

## Failure Cases

Table 4 shows the three row-level outcomes: visual-delta harms, occasional NoLan repairs within aggregate uncertainty, and wrapper scaffolding that leaves retained answers unchanged.

The four visual-delta rows all begin with correct *no* answers and flip to incorrect *yes* answers. They make the aggregate warning concrete: a wrapper can be active, image-conditioned, and directionally harmful at the retained decision boundary. The two NoLan rows show the opposite event—a wrong *no* becomes a correct *yes*—but Table 3 places those repairs beside the method’s harmful changes and uncertainty. They therefore demonstrate that beneficial examples alone cannot establish mitigation.

The two identity controls serve a different purpose. Evidence companion and prompt-label verifier execute their controller paths on the same retained task, yet both preserve the baseline *yes* answer. These rows separate mechanism execution from scorer-visible effect: a valid ledger or diagnostic label is not evidence of mitigation unless it changes the normalized answer correctly. Together, the eight rows make the audit classes inspectable without treating selected examples as prevalence estimates.

This selection does not establish universal wrapper behavior. It is a balanced diagnostic set drawn from the canonical row joins: four harms, two benefits, and two exact-identity controls. Its role is to show how base-identical, non-decisive, parser-sensitive, and harmful outcomes arise in individual records; incidence and external validity remain governed by the complete paired tables and the retained-slice limitations.

These checks guard against two reporting artifacts. Base-identical behavior is established from matched outputs, not rounded aggregate equality. Harm is established from correct-to-wrong flips as well as a lower mean. The retained records support both tests, but they do not reveal why a controller failed internally or how tuning would change it. Repair targets therefore remain separate: the verifier needs a decision-changing signal, the companion needs complete conflict evidence, NoLan needs stable paired movement, and visual-delta needs safer answer-token direction.

## Diagnostic Interpretation

The retained audit uses a simple decision sequence. First, the wrapper must change the normalized answer rather than only add scaffolding around the same yes/no decision. Second, changed rows must be beneficial more often than harmful under the retained scorer. Third, the direction must survive the parser sensitivity check when raw answers are not clean binary outputs. This sequence explains why one score table can contain exact no-ops, non-decisive movement, and a significant degradation without reducing all wrappers to one conclusion.

The active wrappers fail for different reasons. NoLan fixed-alpha changes 40 of 900 paired rows, so its failure is not lack of movement. Its retained deltas remain within paired uncertainty, and the HallusionBench cell is parser-sensitive. Visual-delta contrast changes 155 of 900 paired rows. That movement is large enough to reveal a directional problem: AMBER is significantly negative, while HallusionBench trends downward without a decisive interval.

The AMBER–HallusionBench comparison shows why change rate cannot substitute for change direction. Table 3 records similar visual-delta activity on the two 300-row slices: 57 AMBER answers change (81.0% identity) and 61 HallusionBench answers change (79.7% identity). Their correctness balances differ. AMBER has 19 beneficial against 38 harmful changes, a net loss of 19 rows and a  $-6.3$ -point interval of  $[-11.2, -1.4]$  that excludes zero. HallusionBench has 25 benefits against 36 harms, a net loss of 11 rows and a  $-3.7$ -point interval of  $[-8.8, +1.4]$  that still includes zero. Thus the decisive distinction is not whether contrast perturbs answers, but whether paired harms dominate by enough to survive uncertainty.

The AMBER type split further narrows that imbalance. Attribute questions are balanced at eight benefits and eight harms, whereas existence contributes nine harms with no benefits and relation contributes 21 harms against 11 benefits. This pattern does not identify a universal causal defect in visual contrast, but it does identify the retained failure surface: answer-token contrast is unsafe where object presence and relations must remain jointly consistent. Future wrapper evaluations should therefore require all three signals together—nontrivial retained-answer change, net beneficial paired direction within each benchmark family, and

Table 4: Eight selected POPE rows contain four visual-delta harms, two NoLan benefits, and two exact-identity controls.

| Example   | Wrapper      | Base | Output | Relation        |
|-----------|--------------|------|--------|-----------------|
| COCO #12  | Visual-delta | no   | yes    | harm            |
| COCO #24  | Visual-delta | no   | yes    | harm            |
| COCO #36  | Visual-delta | no   | yes    | harm            |
| COCO #50  | Visual-delta | no   | yes    | harm            |
| COCO #17  | NoLan        | no   | yes    | benefit         |
| COCO #133 | NoLan        | no   | yes    | benefit         |
| COCO #1   | Companion    | yes  | yes    | exact identical |
| COCO #1   | Verifier     | yes  | yes    | exact identical |

Display labels map to canonical task identifiers in the released row-level records.

stability under strict parsing—rather than treating any one signal as sufficient.

This interpretation is benchmark-specific. POPE can make a high-identity wrapper look harmless because the retained object-existence slice is already easy for the frozen baseline. AMBER is more diagnostic for answer-token contrast because its retained all-accuracy mixes object, attribute, and relation phenomena. HallusionBench contributes visual-language illusion rows, but the retained subset supports only question-level paired claims. A future wrapper can pass the same audit by showing non-identity where its mechanism should act, beneficial paired changes without offsetting harms, score agreement with official summaries, and robustness to stricter raw-answer parsing.

The disaggregated reporting also prevents two misleading readings of a negative study. If the paper reported only aggregate means, the prompt-label verifier and evidence companion rows would look like ordinary ties rather than exact retained-output identities. If the paper reported only changed-row counts, visual-delta contrast would look active without showing that the activity is harmful on AMBER and non-decisive on HallusionBench. Keeping mean score, identity, paired direction, parser sensitivity, and selected row examples in the same narrative makes the result falsifiable: a reader can see which step of the audit each wrapper fails. That is the reason the paper treats the wrappers as diagnostic probes rather than as comparable candidate mitigations.

This conservative reading is also what the retained evidence can support. It does not decide whether a tuned verifier, a different NoLan schedule, or another backend would behave better. It does show that wrapper activity alone is not evidence of faithfulness. The no-op controllers leave normalized answers unchanged. NoLan moves but remains inside paired uncertainty. Visual-delta moves more, yet the decisive movement is harmful. These distinctions specify the checks a future training-free wrapper must pass before making a positive retained-benchmark claim.

## Scope and Limitations

The evidence boundary is narrow by design. The study uses one frozen Qwen/Qwen2.5-VL-7B-Instruct backend, one deterministic run per condition, and selected 300-row slices rather than full benchmark releases or repeated seeds. POPE supplies object-existence rows, AMBER mixes existence, attribute, and relation phenomena, and HallusionBench is retained at question level because the selected sub-

set lacks the original companion rows needed for pair-all decomposition. The results are therefore family-specific evidence against positive mitigation claims for this retained matrix, not a ranking of models, seeds, benchmark families, or all possible wrapper implementations.

The parser and implementation boundaries are equally explicit. Non-binary raw outputs pass through the retained normalizer only after negative and affirmative marker checks, so strict invalid-as-wrong sensitivity accompanies the retained results. The wrappers are diagnostic rather than tuned releases: NoLan uses fixed alpha, the verifier and companion use fixed labels and ledgers without learned thresholds, and visual-delta uses canonical yes/no token scores. A stronger positive claim would require beneficial changes without offsetting harms, stable strict-parser behavior, repeated seeds or backends, and fuller benchmark coverage.

Within that scope, 4,500 scored rows and 3,600 paired rows support a stricter audit than mean scores alone. Re-computed task-level scores reproduce all 15 reported cell means within scorer reporting precision. Combining that agreement with normalized identity and beneficial-versus-harmful changes distinguishes base-identical, non-decisive, and active harmful wrappers without pooling the benchmark families. A future wrapper would pass only by changing answers where intended, producing net beneficial paired movement, and preserving direction under strict parsing.

## Conclusion

This paper evaluates training-free, calibration-free wrappers as failure-mode probes around a frozen multimodal model. Prompt-label verifier and evidence-companion arbitration are base-identical; NoLan fixed-alpha is non-decisive and parser-sensitive on HallusionBench; visual-delta significantly degrades AMBER and decreases HallusionBench within uncertainty. Wrapper activity must therefore be evaluated at the retained-answer level, not inferred from extra prompts, probes, or contrast signals.

The contribution is an evidence standard rather than a new decoding recipe: report answer changes, paired benefits and harms, family-specific effects, and strict-parser sensitivity together. The retained wrappers fail at different steps, exposing no-op, non-decisive, parser-sensitive, and harmful behavior.

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## Reproducibility Checklist

**Scope.** The draft reports selected 300-row POPE, AMBER, and HallusionBench subsets with one retained run per method-by-benchmark cell and Qwen/Qwen2.5-VL-7B-Instruct as the frozen backend.

**Released records.** The replay package contains row-level predictions, scorer summaries, paired correctness records, aggregate paired statistics, wrapper-output summaries, and selected qualitative examples. These record types are sufficient to regenerate the reported means, deltas, identity rates, McNemar tests, confidence intervals, and example labels.

**Metrics.** POPE and AMBER use official-score accuracy variants; HallusionBench uses retained question accuracy because pair-all decomposition is unavailable for the selected subset. Paired comparisons use exact McNemar tests on row correctness.

**Randomness and budget.** The scored matrix contains one full run per condition, no parameter training, no calibration set, and no repeated stochastic seed analysis. Generated-answer paths use deterministic decoding with temperature 0 and at most 128 new tokens; claims are limited to the retained rows and do not generalize to other models, seeds, or full benchmark releases.

**Known caveats.** AMBER’s upstream output rounds aggregate percentages to one decimal place, so its row-level consistency check uses the corresponding reporting tolerance. The binary normalizer has a permissive yes default for unrecognized text after explicit no/not and yes checks. These caveats are part of the claim boundary.

## Appendix: Additional Checks

The appendix connects Tables 2 and 3 to replay evidence. Table 5 records method budgets; Table 6 combines score replay, parser checks, matched-row coverage, and changed-row outcomes.

The fixed budgets make the comparison an audit of pre-defined wrapper behavior rather than tuning. Every answer path is deterministic, and no wrapper uses validation labels or stochastic seed selection. Budget differences therefore describe each interface’s fixed computation; they do not conceal per-benchmark calibration.

The replay checks establish three distinct facts. Agreement in all 15 cells ties the paper-facing means to row records. Complete 3,600-row pairing makes benefits, harms, and McNemar tests comparable within each family. The HallusionBench NoLan cell remains a boundary case: counting nine non-binary answers as wrong reverses its retained direction, so the main text reports parser sensitivity rather than improvement.

Table 5: All five evaluated paths use fixed deterministic budgets.

| Method                | Calls / budget                                          |
|-----------------------|---------------------------------------------------------|
| Frozen prompt         | one image-conditioned greedy answer; at most 128 tokens |
| NoLan fixed-alpha     | alpha=1.0 image/text loop; at most 128 tokens           |
| Prompt-label verifier | baseline answer and seven fixed label probes            |
| Visual-delta contrast | baseline answer and image/text yes-no scores            |
| Evidence companion    | baseline answer and one evidence-ledger call            |

Table 6: Replay checks cover all 15 score cells and 3,600 paired rows; changed-row outcomes separate two no-ops from active wrappers.

| Item                             | Quantity        | Evidence                                                          | Reading      |
|----------------------------------|-----------------|-------------------------------------------------------------------|--------------|
| <b>A. Reproducibility checks</b> |                 |                                                                   |              |
| Score replay                     | 15/15 cells     | Row means agree; maximum absolute difference is 0.0.              | pass         |
| Parser sensitivity               | 9/300 invalid   | HallusionBench NoLan changes from 86.0% retained to 83.3% strict. | sensitive    |
| Paired coverage                  | 3,600 rows      | Twelve comparisons retain matched task rows.                      | complete     |
| <b>B. Wrapper outcomes</b>       |                 |                                                                   |              |
| NoLan                            | 40/900 changed  | 24 beneficial; 16 harmful; 95.6% identity                         | non-decisive |
| Verifier                         | 0/900 changed   | 0 beneficial; 0 harmful; 100.0% identity                          | no-op        |
| Visual-delta                     | 155/900 changed | 67 beneficial; 88 harmful; 82.8% identity                         | AMBER harm   |
| Companion                        | 0/900 changed   | 0 beneficial; 0 harmful; 100.0% identity                          | no-op        |

Movement counts use normalized answers. Table 3 gives paired uncertainty and the harmful AMBER direction; the strict parser counts non-binary outputs as wrong.